MIL-P-14901 (MU)
16 Aug. 1968

# MILITARY SPECIFICATION

## PROJECTILE, 4.2 INCH: TACTICAL CS, XM633 LOADING, ASSEMBLING AND PACKING (FOR CARTRIDGE, TACTICAL CS, XM630)

### 1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of projectile designated as Projectile, 4.2 Inch: Tactical CS, XM633.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-3550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1315

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DRAWINGS

ARMY

8848553 -- Packing and Marking of Pallet for 4.2 Inch Mortar Shell Bodies
9220298 - Projectile, 4.2 Inch: Tactical CS, XM633 Loading Assembly

PUBLICATIONS

ARMY

ET-9220298 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite "49 CFR 71-90 (latest revision)").

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.



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3.2 Projectile assembly.-The projectile assembly shall comply with all requirements specified on Drawing (dwg.) 9220298 and with all the requirements specified in the applicable specifications.

3.3 Consolidation of canister assembly.-The projectile contents shall be tightly packed. To insure a tight pack, a 3/16 inch (minimum) gap is required between the body shoulder and the outer front face of a simulated base plug when the simulated plug is positioned on the spacers in the base of the projectile (see 4.3.3.1) before the final assembly of the base plug proper.

3.4 Proving ground

3.4.1 The projectile loading assembly shall comply with the following requirements when tested as specified in 4.4.2. The requirements of 3.4.2 and 3.4.3 are not applicable when the contractor for the projectile assembly is also the contractor for the assembling and packing of the complete round.

3.4.2 Functioning.-The projectile loading assembly shall function satisfactorily (see 6.5) and the canisters shall burn for 30 seconds minimum.

3.4.3 Security of parts.-There shall be no metal parts failure in the bore of the weapon or in the first approximate 100 yards of horizontal range at any service charge or with propellant adjusted to give a chamber pressure of 17,300 psi (approximately 47 increments).

3.5 Workmanship

3.5.1 All parts and assemblies shall be fabricated in a thorough workmanlike manner. They shall be free of burrs, irregularities, chips, sharp edges, cracks, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.5.2 The painting and marking shall be restored or repainted by the assembling contractor on any components furnished to him painted and marked, when such paintings and markings have been marred, obliterated or require color changes.



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3.6 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.



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4.2 First Article Inspection

4.2.1 The contractor, prior to initiation of regular production, shall submit a first article sample to a Government approved facility as designated by the contracting officer (see 6.4) for evaluation for conformity to drawing and specification requirement. The first article sample shall consist of inert parts and sub-assemblies (except canister metal parts), and loaded projectiles which have been produced by the contractor, or furnished by a supplier, and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by a statement of findings for materials. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever a lapse occurs in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.1.1 Inert parts and sub-assemblies

4.2.1.1.1 Submission.-Fifteen complete sets of parts (i.e. 15 each of every inert component and inert sub-assembly) shall be submitted for inspection to a Government approved laboratory designated by the contracting officer (see 6.4).

4.2.1.1.2 Inspections to be performed.-The inert components and sub-assemblies shall be inspected for conformance with any or all of the requirements of this specification and any or all of the requirements of the applicable drawings.

4.2.1.1.3 Rejection.-If any inert component or sub-assembly fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any inert component or sub-assembly in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample or portion thereof of inert components or sub-assemblies for first article inspection.



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4.2.1.2 Loaded projectiles

4.2.1.2.1 Submission.-Twenty loaded projectiles shall be inspected for conformance to any or all of the requirements of this specification and any or all of the requirements of the applicable drawings at the contractor's plant prior to submission to the proving ground for proving ground testing. The proving ground test shall be performed in accordance with 4.3.3.2.

4.2.1.2.2 Rejection.-If any projectile fails to comply with any of the requirements specified, the initial production sample shall be rejected. The Government reserves the right to terminate testing upon any failure of a projectile in the sample to comply with the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample or portion thereof of loaded projectiles for first article testing. Until that phase of the first article sample consisting of loaded projectiles is ballistically accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection Provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specification, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of



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delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of projectile loading assemblies shall contain:

a. Metal parts and metal parts assemblies from one lot interfix number from one manufacturer.
b. Canisters from one lot from one manufacturer.
c. Projectiles of one weight zone.

4.3.2 Examination.-Inspection for Critical defects (and Major defects, when so specified), shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Baffle plate (see dwg. 9234360 covering a detail of dwg. 9220298).

| Categories | Defects | Method of Inspection | Code No. (see 6.2) |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Diameter of baffle plate ..... | Gage | 01001 |
| 102. | Thickness, minimum (min.) ..... | Gage | 01002 |
| 103. | Diameter of hole, (min.) ..... | Gage | 01003 |
| 104. | Concentricity of hole ..... | Gage | 01004 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.5) ..... | Visual | 01005 |

4.3.2.2 Base Filler (see dwg. 9234359 covering a detail of dwg. 9220298).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |



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| | | | |
|---|---|---|---|
| Major: | AQL 0.65 percent | | |
| 101. | Diameter, (min.) ..... | Gage | 02001 |
| 102. | Side curvature, (min.) ..... | Gage | 02002 |
| 103. | Thickness, (min.) ..... | Gage | 02003 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.5) ..... | Visual | 02004 |

4.3.2.3 Projectile prior to assembly of base plug (see dwg. 9220298).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.0 percent | | |
| 101. | Baffle plate missing or improperly assembled ..... | Visual | 03001 |
| 102. | Canister improperly assembled or damaged to the extent that contents will escape ..... | Visual | 03002 |
| 103. | Canister assembled with crimped end not facing nose ..... | Visual | 03003 |
| 104. | Canister spacer(s) missing ..... | Visual | 03004 |
| 105. | Base filler missing or loose ... | Visual-Manual | 03005 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.5) ..... | Visual | 03006 |

4.3.2.4 Projectile loading assembly (see dwg. 9220298).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Shear pin(s) missing..... | Gage | 04001 |
| Major: | AQL 1.0 percent | | |
| 101. | Projectile weight zone incorrect ..... | Scale | 04002 |
| 102. | Zone stenciling incorrect ..... | Visual | 04003 |
| 103. | Excessive gap between base plug and body ..... | Gage | 04004 |
| 104. | Twist pin(s) missing ..... | Visual | 04005 |



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| | | | |
|---|---|---|---|
| 105. | Base plug threads obviously damaged | Visual | 04006 |
| 106. | Pin(s) above flush with outer surface | Visual-Manual | 04007 |
| 107. | Shear pin(s) or twist pin (s) not staked in place | Visual | 04008 |
| 108. | Diameter of bourrelet, (max.) | Gage | 04009 |

Minor: AQL 1.5 percent

| | | | |
|---|---|---|---|
| 201. | Marking misleading, missing or unidentifiable (except for zones), including improper color | Visual | 04010 |
| 202. | Nose closing plug missing | Visual | 04011 |
| 203. | Exterior coating of projectile with an aggregate of bare spots in excess of 1/4 square inch | Visual | 04012 |
| 204. | Evidence of poor workmanship (see 3.5) | Visual | 04013 |

4.3.2.5 Pallet, prior to strapping (see dwg. 8848553).

| Categproes | Defects | Method of Inspection | Code No. |
|---|---|---|---|

Critical: None defined.

Major: None defined.

Minor: AQL 1.5 percent

| | | | |
|---|---|---|---|
| 201. | Liner missing | Visual | 05001 |
| 202. | Sleeve missing | Visual | 05002 |
| 203. | Evidence of poor workmanship (see 3.5) | Visual | 05003 |

4.3.2.6 Pallet (see dwg. 8848553).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|

Critical: None defined

Major: AQL 0.65 percent

| | | | |
|---|---|---|---|
| 101. | Strapping loose | Visual-Manual | 06001 |
| 102. | Boards broken or missing | Visual | 06002 |



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| | | | |
|---|---|---|---|
| 103. | Projectiles of different weight zones on the same pallet | Visual | 06003 |
| Minor: | AQL 1.0 percent | | |
| 201. | Marking, misleading or unidentifiable | Visual | 06004 |
| 202. | Evidence of poor workmanship (see 3.5) | Visual | 06005 |

4.3.3 Testing

4.3.3.1 Consolidation of canister assembly - Major defect - Code No. 3007.-The projectiles shall be gaged 100 percent for this test. The test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4.

4.3.3.2 First Article Inspection

4.3.3.2.1 Functioning.-Fifteen samples of the twenty sample projectiles shall be fired in three groups of five rounds each at propelling charge levels of 12, 25 1/2 and 41 increments. Observation shall be made for conformance to the applicable requirements (see 3.4 and Table 1). In addition, the proving ground shall make the necessary arrangements to recover any or all malfunctioning canisters and rounds which may occur during the test. Arrangements for the recovery of malfunctions shall be for the first article samples only unless otherwise directed by the procuring activity.(see 4.4.2.1).

4.3.3.2.2 Safety.-Five samples of the twenty sample projectiles shall be fired for this test. Observation shall be made for conformance to the applicable requirement (see 3.4.3 and 4.3.3.4). The test shall be performed as specified in 4.4.2.2.

4.3.3.3 Production

4.3.3.3.1 Functioning

4.3.3.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified in 3.4 forty-eight sample projectiles shall be selected for this test. The projectiles shall be fired in three groups of sixteen rounds each at propelling charge levels of 12, 25 1/2 and 41 increments. The projectiles shall be observed for compliance with the requirements specified in Table I (see 4.4.2.1).



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4.3.3.3.1.2 After three consecutive lots have met the criteria of 4.3.3.3.1.1, fifteen sample projectiles shall be selected for this test. The projectiles shall be fired in three groups of five rounds each at propelling charge levels of 12, 25 1/2 and 41 increments. The projectiles shall be observed for compliance with the characteristics specified in Table I (see 4.4.2.1).

TABLE I

| Category | Defect | Sample Size 48 (192 Canisters) (see 4.3.3.3.1.1) Acc. | Rej. | Sample Size 15 (60 Canisters) (see 4.3.3.2.1 and 4.3.3.3.1.2) Acc. | Rej. | Code No. |
|---|---|---|---|---|---|---|
| Critical | Security of parts (see 4.3.3.4) | 0 | 1 | 0 | 1 | 04014 |
| Major* | Canisters do not burn for 30 seconds | 12 | 13 | 4 | 5 | 04015 |

*NOTE: Pertains to number of canisters (four canisters per projectile).

4.3.3.3.2 Safety

4.3.3.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements (see 4.3.3.4), 5 sample projectiles shall be selected for this test. If any projectile fails to comply with the "security of parts" requirements, the lot shall be rejected. The test shall be performed as specified in 4.4.2.2.

4.3.3.3.2.2 After three consecutive lots have met the criteria of 4.3.3.3.2.1, three sample projectiles shall be selected for this test. If any projectile fails to comply with the "security of parts requirements" (see 4.3.3.4), the lot shall be rejected. The test shall be performed as specified in 4.4.2.2.

4.3.3.4 Security of parts (see 3.4.3) - Critical defect - Code No. 04014.-If any of the sample projectiles fail to comply with the applicable requirement regarding metal parts failure, the lot shall be rejected. Any premature burst that is attributed to premature fuze functioning shall not be cause for rejection of the projectile lot.



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4.3.3.5 Refiring.-If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional sample projectiles as required.

4.3.4 Inspection Equipment.-Index of Inspection Equipment List Number ET-9220293 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Consolidation of canister assemblies.-A simulated base plug having the dimensions of the base plug (see dwg. 10534160) except that the inner section diameter of 3.814-.001 inches shall be sized to permit a free fit between this part of the simulated base plug and the projectile body. The simulated base plug shall be inserted into the projectile body and positioned on the base spacers. A minimum gap of 3/16 inch shall exist between face of the simulated base plug and the rear shoulder of the projectile body. If the gap does not meet this requirement, additional spacers are required.



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4.4.2 Proving ground

4.4.2.1 Functioning.-The projectile assemblies shall be assembled with accepted fuzes, expelling charges, propellant, ignition cartridges and obturating mechanisms. All of the cartridges shall be fired from a 4.2 inch mortar for which the round is standard and in accordance with Table II. However, if meterological conditions or time of fuze functioning so dictate, the proving ground shall adjust the fuze in order to preclude the rounds from bursting on the ground. Observation shall be made for compliance with the requirements specified for security of parts and canister burning time. Muzzle velocity shall be recorded for information only for each round tested. Any malfunction occurring during the test shall be reported in the firing record. Meterological data shall also be reported in the official firing record. In addition for the first article sample only, a smear camera shall be used to observe the projectile in flight at approximately 25 feet in front of the weapon, in order to record any possible premature malfunctions.

*TABLE II

| Propellant Phase | Fuze Setting (sec.) | Mortar Elevation (mils) |
|---|---|---|
| 12 | 11.0 | 900 |
| 25 1/2 | 28.0 | 800 |
| 41 | 36.0 | 800 |

*Above fuze setting and elevations are nominal.

4.4.2.2 Safety.-The projectile for this test shall be assembled with the propellant adjusted to give a chamber pressure of 17,300 psi (approximately 47 increments) and fired at an elevation of 900 mils with the fuze set at maximum time. Observation shall be made for premature bursts and indication of metal parts failure in the gun bore or in flight. In addition, for the first article sample only, a smear camera shall be used to observe the projectile in flight at approximately 25 feet in front of the weapon, in order to record any possible premature malfunctions.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level C.-The loaded projectiles shall be packed in accordance with dwg. 8848553.



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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing the the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor



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shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NDL. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.



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6.4 Submission of first article samples.-Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer. The contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI, Dover, New Jersey 07801.

6.5. Satisfactory functioning.-Satisfactory functioning shall be defined as burning of the canister for the required time without detonation. (Detonator is disintegration of the canister assembly in the air after ejection from the projectile which results in a puff of smoke or dust cloud).

6.6 Submission of test data.-In addition to the normal distribution of records, when the projectile is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND2 and DA3
Dover New Jersey 07801

Custodian:
Army-MU

Preparing Activity
Army-MU

Project Number: 1315-A-596



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

| ORGANIZATION | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

DD FORM 1 OCT 64 1426

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

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DEPARTMENT OF THE ARMY

COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

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343-220 (A-329)

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MILITARY SPECIFICATION

# PROJECTILE, 4.2 INCH: TACTICAL CS, XM633 LOADING ASSEMBLING AND PACKING (FOR CARTRIDGE, TACTICAL, XM630)

This amendment forms a part of Military Specification MIL-P-14901 (MU), dated 16 August 1968.

Page 1

Add the following Preamble underneath the title: "This specification is approved by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense."

Page 3

3.3, line 3: Delete "3/16 inch" and substitute "1/16 inch."

Page 6

4.3.1: Delete sentence "b" in its entirety and substitute the following: "b. Canisters from one lot interfix number from one manufacturer, except not more than three canister lots shall be used in any lot of loaded projectiles."

Page 8

4.3.2.3: Add new Major Defect as follows: "106. Shipping tape not removed from each end of canister....Visual 03008."

Page 12

4.4.1, line 7: Delete "3/16 inch" and substitute "1/16 inch."

Custodian:
Army - AR

Preparing Activity:
Army - AR

Project No: 1315-A434

FSC 1315

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NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH-POUND

MIL-P-14901(AR)
NOTICE 2
31 January 1996

MILITARY SPECIFICATION

# PROJECTILE, 4.2 INCH: TACTICAL CS, XM633 LOADING, ASSEMBLING AND PACKING (FOR CARTRIDGE, TACTICAL CS, XM630)

This notice should be filed in front of MIL-P-14901(AR), dated 16 August 1968

MIL-P-14901(AR), dated 16 August 1968, is inactive for new design and is no longer used , except for replacement purposes.

Preparing Activity:
Army - AR

AMSC N/A FSC 1315

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